Forever FitBand

An offline, privacy-preserving AI health wearable with personal disaster early warning

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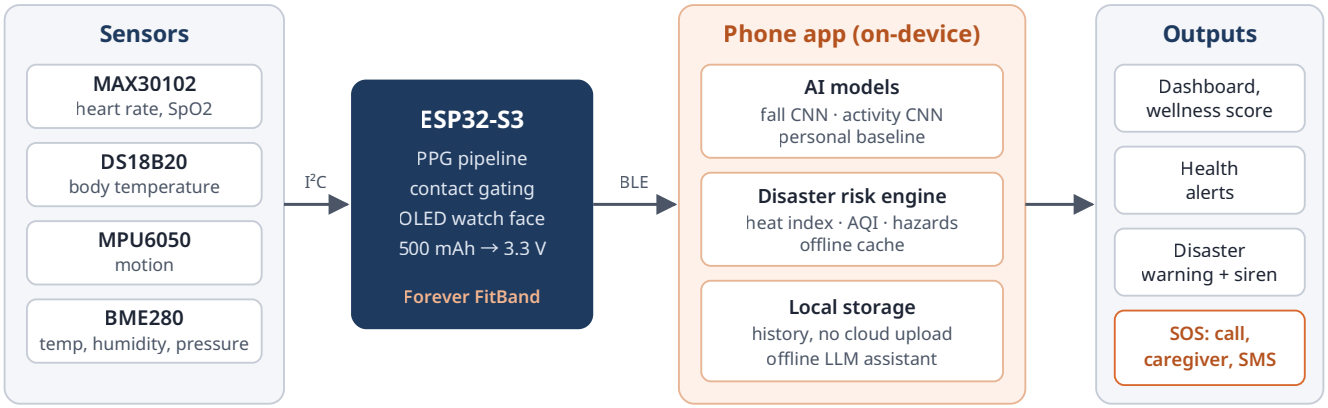
PROBLEM

During heat waves, floods, pollution events and earthquakes, people get generic alerts but no warning that conditions are dangerous *for them*. Most wearables depend on the cloud and stop helping when networks fail, and a fall at home can go unnoticed.

METHODOLOGY

1. **Sense:** an ESP32-S3 wrist band reads heart rate and SpO2 (MAX30102), body temperature (DS18B20), motion (MPU6050) and ambient conditions (BME280), powered by a 500 mAh battery.
2. **Validate:** the PPG pipeline settles for 2 s, rejects outlier beats and reports only after 4 clean beats; skin-contact detection blocks false readings.
3. **Transmit:** vitals, environment and motion stream to the phone over Bluetooth Low Energy.
4. **Infer on-device:** a fall-detection CNN, an activity CNN that sets context-aware heart-rate limits, and a 7-day personal baseline flag anomalies. A risk engine fuses vitals with heat index, air quality and hazard data, cached for offline use.
5. **Act:** alerts, full-screen disaster warnings with a siren, and SOS escalation after a 10 s cancel window.

PROPOSED BLOCK DIAGRAM



PROPOSED OUTCOME

- A working wearable and app that monitor vitals continuously and **keep working with no internet**.
- Automatic fall detection (92% precision and recall on unseen test subjects) with emergency escalation.
- Personalised early warnings for heat, air quality, floods, cyclones and earthquakes.
- Complete privacy: health data never leaves the user's phone.
- A low-cost design (about ₹2,200 in parts) suited to elderly and at-risk users.

Demo: youtu.be/iUX2lkdWSqk · Source: GitLab